

① KKT Conditions:-a) write the objective function $f(x)$ & constraints $g_i(x) \leq 0$

b) From the lagrangian function:-

$$L = f(x) + \sum \lambda_i g_i$$

$$\frac{\partial L}{\partial x_j} = 0$$

• stationarity

• primal feasibility $g_i(x) \leq 0$ • Dual Feasibility $\lambda_i \geq 0$ • Complementary slackness $\lambda_i g_i(x) = 0$

c) solve equations simultaneously

d) check feasibility & compute optimal value.

② Beale's method:-

$$f(x) = 4x_1 + 6x_2 - x_1^2 - 3x_2^2$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

S₁:- From lagrangian,

$$L = 4x_1 + 6x_2 - x_1^2 - 3x_2^2 + \lambda (4 - x_1 - 2x_2)$$

S₂:- Partial derivation:-

$$\frac{\partial L}{\partial x_1} = 4 - 2x_1 - \lambda = 0$$

S₃:- solve

$$\lambda = 4 - 2x_1$$

substitute λ .

$$6 - 6x - 2(4 - 2x_1) = 0$$

$$6 - 6x_2 - 8 + 4x_1 = 0$$

$$4x_1 - 6x_2 - 2 = 0$$

S4:- $x_1 + 2x_2 = 4$

Solve $x_1 = 2$ & $x_2 = 1$.

S5:- $Z = 8 + 6 - 4 - 3 = 7$

optimal solution $(2, 1)$:- $Z_{\max} = 7$.

③ Wolfe's method:-

$$Z_{\max} = 3x_1 + 4x_2$$

$$x_1^2 + x_2^2 \leq 4$$

S1:- Lagrangian

$$L = 3x_1 + 4x_2 + \lambda(4 - x_1^2)$$

S2:- derivatives :-

$$3 - 2\lambda x_1 = 0 \rightarrow x_1 = \frac{3}{2\lambda}$$

$$4 - 2\lambda x_2 = 0 \rightarrow x_2 = \frac{2}{\lambda}$$

S3:- substitute

$$\left(\frac{3}{2\lambda}\right)^2 + \left(\frac{2}{\lambda}\right)^2 = 4$$

$$\frac{9}{4\lambda^2} + \frac{4}{\lambda^2} = 4$$

$$\boxed{\lambda = \frac{5}{4}}$$

S4:- values

$$x_1 = \frac{3}{2 \times \frac{5}{4}} = \frac{6}{5}$$

$$x_2 = \frac{8}{5}$$

S5:- Z value

$$Z = \frac{18}{5} + \frac{32}{5}$$

$$Z = 10$$

optimal solution

$$(6/5, 8/5) \quad Z_{\max} = 10$$

$$H) \quad Z_{\min} = 2x_1 + x_2 + 3x_3 + 10x_1 + 8x_2 + 6x_3 - 100$$

$$x_1 + x_2 + x_3 = 20$$

S1:- Lagrangian

$$L = Z + \lambda(20 - x_1 - x_2 - x_3)$$

S2:- Derivatives

$$4x_1 + 10 - \lambda = 0$$

$$2x_2 + \frac{8}{\lambda} = 0$$

$$6x_3 + 6 - \lambda = 0$$

S3:- solving from all;

$$\lambda = 4x_1 + 10 = 2x_2 + 8 = 6x_3 + 6$$

$$x_1 = 2; \quad x_2 = 4; \quad x_3 = 14$$

S4:- values

$$Z = -2 (\min)$$

$$\text{max at } (2, 4, 14) \quad Z_{\min} = -2$$

(E) KKT problem. $\Rightarrow f = 4x_1 + 3x_2 - x_1^2 - x_2^2 = 0$

$$\text{sol) } L = f + \lambda(2 - x_1 - 2x_2) \quad x_1 + 2x_2 \leq 2$$

Derivatives:-

$$4 - 2x_1 - \lambda = 0$$

$$3 - 2x_2 - \frac{\lambda}{2} = 0$$

By solving,

$$x_1 = 1, \quad x_2 = 0.5$$

check the constraints:-

$$1 + 1 = 2$$

values

$$Z = 4 + 1.5 - 1 - 0.25 \\ = 4.25$$

optimal solution is $(1, 0.5) = 4.25$

⑥ Geometric Method:-

$$y_1 = x_1^{-1}, y_2 = x_2^{-1}, y_3 = x_3^{-1}$$

① Transfer into simplex form

Convert non-linear terms $\rightarrow$ linear in logs.

② Apply geometric programming:-

- Express as sum of constraints
- Use dual formulation.
- solve using equality conditions.

③ final:-

$$x_1 = 1, x_2 = 1, x_3 = 1$$

optimal solution is $(1, 1, 1)$.

~~24/3~~
24/3